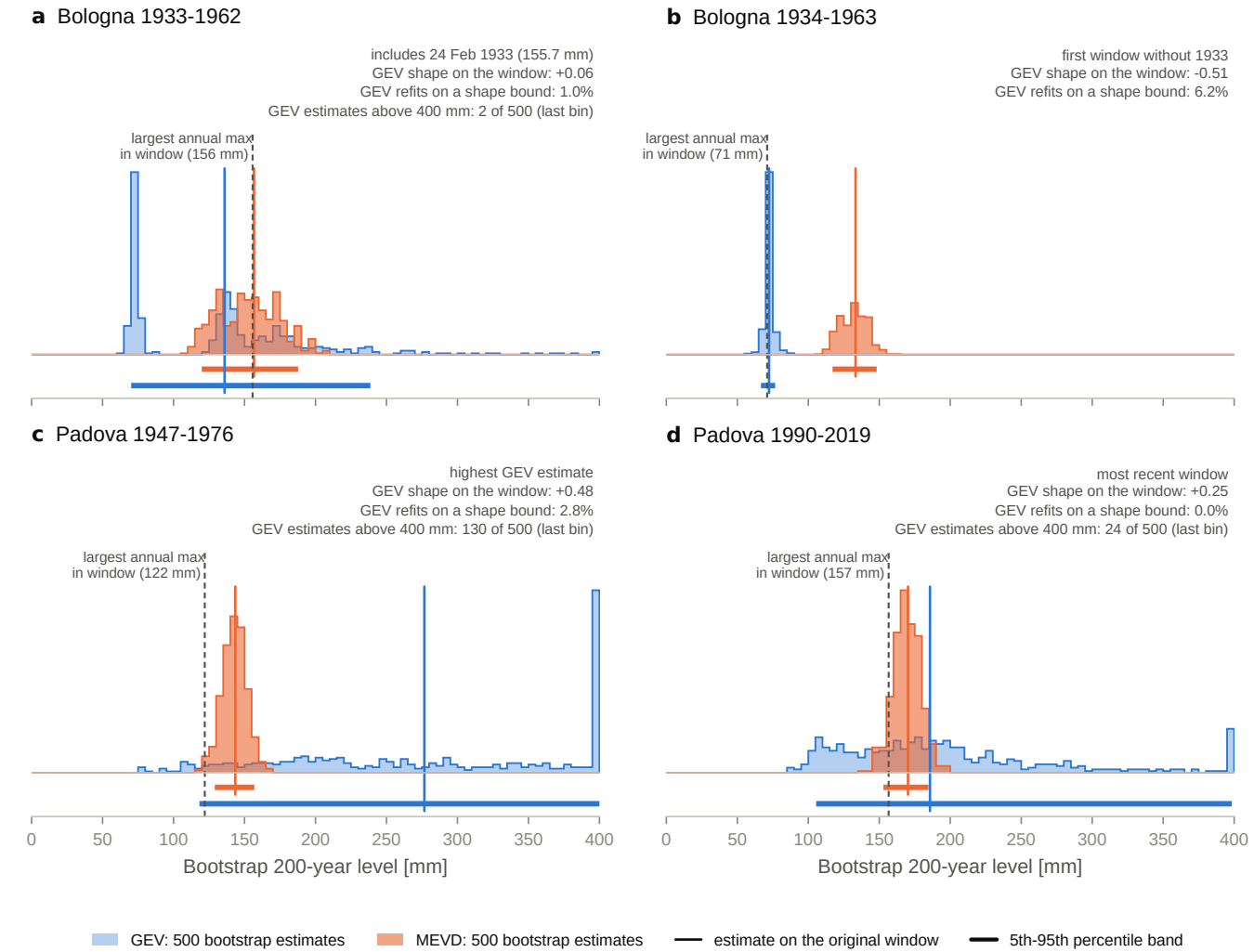What the bands are made of: bootstrap distributions in selected windows